Description

Image



Caption

1. A GFRP component being made by spraying resin onto fiberglass. 2. A swimming pool made from GFRP. 3. Large diameter GFRP pipes for sewer construction. Images used under license from Shutterstock.com

The material

Composites are one of the great material developments of the 20th century. Those with the highest stiffness and strength are made of continuous fibers (glass, carbon or Kevlar, an aramid) embedded in a thermosetting resin (polyester or epoxy). The fibers carry the mechanical loads, while the matrix material transmits loads to the fibers and provides ductility and toughness as well as protecting the fibers from damage caused by handling or the environment. It is the matrix material that limits the service temperature and processing conditions. Polyester-glass composites (GFRPs) are the cheapest and by far the most widely used. A recent innovation is the use of thermoplastics as the matrix material, either in the form of a co-weave of cheap polypropylene and glass fibers that is thermoformed, melting the PP, or as expensive high-temperature thermoplastic resins such as PEEK that allow composites with higher temperature and impact resistance. High performance GFRP uses continuous fibers. Those with chopped glass fibers are cheaper and are used in far larger quantities. GFRP products range from tiny electronic circuit boards to large boat hulls, body and interior panels of cars, household appliances, furniture and fittings.

Compositional summary

Epoxy + continuous E-glass fiber reinforcement (0, +-45, 90), quasi-isotropic layup.

General properties

Density	1.75e3	-	1.97e3	kg/m^3
Price	* 25.4	-	28	GBP/kg
Date first used	1935			

Mechanical properties

Young's modulus	* 21	-	21.8	GPa
Shear modulus	* 7.99	-	8.29	GPa

Bulk modulus	*	6.8	-	7.85	GPa
Poisson's ratio	*	0.314	-	0.315	
Yield strength (elastic limit)	*	207	-	304	MPa
Tensile strength	*	207	-	304	MPa
Compressive strength	*	207	-	257	MPa
Elongation	*	0.85	-	0.95	% strain
Hardness - Vickers	*	10.8	-	21.5	HV
Fatigue strength at 10^7 cycles	*	41.3	-	91.1	MPa
Fracture toughness	*	19.3	-	31	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0028	-	0.0029	

Thermal properties

Glass temperature		99.9	-	180	°C
Maximum service temperature	*	140	-	220	°C
Minimum service temperature	*	-123	-	-73.2	°C
Thermal conductor or insulator?		Poor insulator			
Thermal conductivity	*	0.42	-	0.51	W/m.°C
Specific heat capacity	*	1.02e3	-	1.12e3	J/kg.°C
Thermal expansion coefficient	*	8.64	-	33	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity	*	2.4e21	-	1.91e22	µohm.cm
Dielectric constant (relative permittivity)	*	4.86	-	5.17	
Dissipation factor (dielectric loss tangent)	*	0.00153	-	0.00697	
Dielectric strength (dielectric breakdown)	*	11.8	-	19.7	MV/m

Optical properties

Transparency		Translucent			

Critical Materials Risk

High critical material risk?		No			

Processability

Moldability		4	-	5	
Machinability		2	-	3	

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			

Soils, acidic (peat)	Limited use
Soils, alkaline (clay)	Limited use
Wine	Limited use

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Limited use
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Acceptable
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Limited use

Durability: alkalis

Sodium hydroxide (10%)	Limited use
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Acceptable
Benzene	Acceptable
Carbon tetrachloride	Excellent
Chloroform	Unacceptable
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Acceptable
Turpentine	Acceptable
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Unacceptable
Acetone	Unacceptable
Ethyl alcohol (ethanol)	Limited use
Ethylene glycol	Acceptable
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Unacceptable

Durability: halogens and gases

Chlorine gas (dry)	Limited use
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Fair

Durability: flammability

Flammability	Slow-burning

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Limited use
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	*	6.22	-	6.86	kg/kg
Embodied energy (virgin grade production)	*	101	-	111	MJ/kg
Water usage	*	192	-	213	l/kg

Material processing: energy

Autoclave molding, embodied energy	*	70.3	-	77.5	MJ/kg
Compression molding, embodied energy	*	10.1	-	11.1	MJ/kg
Filament winding, embodied energy	*	7.72	-	8.51	MJ/kg

Resin spray-up, embodied energy	* 43.5	-	48	MJ/kg
Resin transfer molding (RTM), embodied energy	* 34.4	-	37.9	MJ/kg

Material processing: climate change

Autoclave molding, climate change (CO2-eq)	* 4.62	-	5.1	kg/kg
Compression molding, climate change (CO2-eq)	* 0.662	-	0.731	kg/kg
Filament winding, climate change (CO2-eq)	* 0.507	-	0.56	kg/kg
Resin spray-up, climate change (CO2-eq)	* 2.85	-	3.14	kg/kg
Resin transfer molding (RTM), climate change (CO2-eq)	* 2.26	-	2.5	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 4.98	-	5.49	kg/kg
Embodied energy, recycling	* 80.8	-	88.8	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 12	-	12.6	MJ/kg
Combustion CO2	* 0.968	-	1.02	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Fiber composites cannot be recycled.

Supporting information

Design guidelines

Polymer composites can be formed by closed or open mold methods. All the closed mold methods produce fiber orientation parallel to the mold surfaces (for extrusion, it is parallel to the inside surface of the orifice die). Of the open mold methods, all allow multidirectional fiber orientation parallel to the mold or mandrel, except pultrusion, where the fibers are oriented parallel to the laminate surface and the mold plates, and calendaring, where they are parallel to the sheet surface. Lay up methods allow complete control of fiber orientation; they are used for large one-off products that do not require a high fiber-resin ratio. Lamination and calendaring form sheets, pultrusion is used to make continuous shapes of constant cross section and filament winding produces large hollow items such as tubes, drums or other containers. Joints in long-fiber composite materials are sources of weakness because the fibers do not bridge the joint. Two or more laminates are usually joined using adhesives and, to ensure adequate bonding, an overlap length of 25mm for single- and double- lap joints or 40-50mm for strap, step and scarf joints is necessary. Holes in laminates dramatically reduce the failure strength making joining with fasteners difficult. Composite manufacture is labor intensive. It is difficult to predict the final strength and failure mode because defects are easy to create and hard to detect or repair.

Technical notes

The properties of long fiber composites are strongly influenced by the choice of fiber and matrix and the way in which these are combined: fiber-resin ratio, fiber length, fiber orientation, laminate thickness and the presence of fiber/resin coupling agents to improve bonding. Glass offers high strength at low cost; carbon has very high strength, stiffness and low density; Kevlar has high strength and low density, is flame retardant and transparent to radio waves (unlike carbon). Polyesters are the most widely used matrices as they offer reasonable properties at relatively low cost. The superior properties of epoxies and the temperature performance of polyimides can justify their use in certain applications, but they are expensive. The strength of a composite is increased by raising the fiber-resin ratio, and orienting the fibers parallel to the loading direction. The longer the fibers, the more efficient is the reinforcement at carrying the applied loads, but shorter fibers are easier to process and hence cheaper. Increased laminate thickness leads to reduced composite strength and modulus as there is an increased likelihood of entrapped voids. Coupling agents generally increase tensile strength. Environmental conditions affect the performance of composites: fatigue loading, moisture and heat all reduce allowable strength.

Typical uses

Sports equipment such as skis, racquets, skate boards, golf club shafts, ship and boat hulls, body shells, automobile components, cladding and fittings in construction, chemical plants.

Tradenames

Cycom, Fiberdux, Scotchply

Links

ProcessUniverse

Producers

Reference